Review HP:0010876 Abnormal circulating protein concentration

Parent class: Abnormal circulating protein concentration (→ „Stammbaumansicht“)

- Alle direkten Subklassen ab HP:0012116 Abnormal circulating albumin concentration bis HP:0031031 Abnormal retinol-binding protein level sind hier am richtigen Platz (ok).
- Über HP:0005559 Abnormality of the kinin-kallikrein system hatten wir bereits gesprochen (komplexes System, bei dem verschiedene Proteine/Enzyme zusammenwirken, siehe Kommentar Tabelle HPO Chemical Phenotypes). Diese Klasse ist hier formal am richtigen Platz.
- Alle übrigen Klassen ab HP:0025498 Aceruloplasminemia bis HP:0031419 Reduced sex-hormone binding protein level gehören nicht in ihr aktuelles Level. Diese Klassen sollte eine neue Parent Class gemäß der Vorlage „Abnormal circulating … concentration“ erhalten und entsprechend als Subklasse darunter eingeordnet werden.

Parent class: Abnormal circulating … concentration

- Alle Klassen **ab HP:0012116 Abnormal circulating albumin concentration bis HP:0025498 Aceruloplasminemia** habe ich im Einzelnen darauf gecheckt, ob sie die entsprechenden Subklassen „Elevated circulating … concentration“ und „Decreased circulating … concentration“ (resp. entsprechende äquivalente medizinische Termini) enthalten.

- Folgende Subklassen waren hier fehlerhaft resp. unvollständig:

HPO-ID	Label	Comment	to do
HP:0006254	Elevated circulating alpha-fetoprotein concentration	I suggest adding HP:0011432 Elevated maternal circulating alpha-fetoprotein concentration as subclass to HP:0006254 Elevated circulating alpha-fetoprotein concentration.	add
HP:0004639	Elevated amniotic fluid alpha-fetoprotein	This class refers to elevated alpha-fetoprotein in amniotic fluid (**not in blood**) and therefore should be eliminated here.	eliminate
HP:0045057	Decreased circulating alpha-fetoprotein concentration	I suggest adding HP:0010570 Low maternal circulating alpha-fetoprotein concentration (new label: "Decreased maternal circulating alpha-fetoprotein concentration") as subclass to HP:0045057 Decreased circulating alpha-fetoprotein concentration.	add
HP:6000566	Elevated circulating lipoprotein X concentration	Lipoprotein-X is not an apolipoprotein, but a lipoprotein (apolipoprotein refers to the protein component of lipoproteins that circulate in the blood). Lipoprotein X (LPX) is an abnormal lipoprotein, recognized by the presence of apoA-I, apoE, and apoC, and lack of apoB.	eliminate
HP:0031138	Abnormal circulating B-type natriuretic peptide concentration	Subclass "Decreased circulating B-type natriuretic peptide concentration" is missing (I suggest the label: "Decreased circulating BNP concentration"). The labels of HP:0031138 and HP:0033534 should be changed accordingly: "Abnormal/Elevated circulating BNP concentration". Furthermore, the subclass "Decreased circulating NT-proBNP concentration" is missing.	add
HP:0011021	Abnormal circulating enzyme concentration	This class is very comprehensive and therefore problematic. Other proteins also belong to this group that are not listed here, e.g., amylase. Does it make sense to retain this class? Or might it be sensible to classify the relevant enzymes/proteins as direct subclasses to HP:0010876 Abnormal circulating protein concentration? Otherwise, every protein listed in HP:0010876 Abnormal circulating protein concentration should be examined to determine whether it is also an enzyme and has to be classified accordingly as subclass to HP:0011021 Abnormal circulating enzyme concentration. **Note**: See also HP:0012379 Abnormal circulating enzyme concentration or activity (142 subclasses). From my point, HP:0012379 and HP:0011021should be compared and adapted if necessary.	check

HP:0040081	Abnormal circulating creatine kinase concentration	Subclass “Decreased circulating creatine kinase BB isoform concentration” is missing (I suggest the label: “Decreased circulating CK-BB concentration”). The labels of HP:0040081 and HP:0003236 should be changed accordingly to “Abnormal/Elevated circulating CK concentration” and the label of HP:0032233 to “Elevated circulating CK-BB concentration”.	add/ rename
HP:0040081	Abnormal circulating creatine kinase concentration	Subclass “Decreased circulating creatine kinase MB isoform concentration” is missing (I suggest the label: “Decreased circulating CK-MB concentration”). The label of HP:0032232 should be changed accordingly to “Elevated circulating CK-MB concentration”.	add/ rename
HP:0040081	Abnormal circulating creatine kinase concentration	Subclass “Decreased circulating creatine kinase MM isoform concentration” is missing (I suggest the label: “Decreased circulating CK-MM concentration”). The label of HP:0032234 should be changed accordingly: “Elevated circulating CK-MM concentration”.	add/ rename
HP:0040081	Abnormal circulating creatine kinase concentration	Note: It might be sensible to classify the subclasses elevated/decreased circulating CK-MB/CK-MM/CK-BB concentration as subclasses to the corresponding classes elevated/decreased circulating creatine kinase concentration, i.e., “Decreased circulating CK-MM concentration” as subclass to “Decreased circulating creatine kinase concentration”.	check
HP:6000213	Elevated circulating Angiotensin-converting enzyme concentration	Parent class “Abnormal circulating Angiotensin-converting enzyme concentration” and sibling class “Decreased circulating Angiotensin-converting enzyme concentration” are missing (I suggest the labels: “Abnormal/elevated/decreased circulating ACE concentration”).	Add/ rename
HP:0003333	Increased serum beta-hexosaminidase	Parent class “Abnormal circulating serum beta-hexosaminidase concentration” and sibling class “Decreased circulating serum beta-hexosaminidase concentration” are missing. The label of HP:0003333 should be adapted as follows: “Elevated circulating beta-hexosaminidase concentration”.	Add/ rename
HP:0032179	Abnormal circulating globulin concentration	This class is very comprehensive and therefore problematic. 40 % of all proteins in the blood are globulins. They are categorized into alpha-, beta-, and gamma-globulins. Some proteins belong to this group that are not listed here, e.g., alpha-1-antitrypsin.	check/add/ reclassify

		If the parent class HP:0032179 is to be retained, it would certainly make sense to make a complete subdivision into the three classes "Abnormal circulating alpha-/beta-/gamma-globulin concentration" with the corresponding (6) subclasses "Elevated/Decreased circulating alpha-/beta-/gamma-globulin concentration". The two classes HP:0025465 Abnormal circulating beta globulin level and HP:0005413 Increased alpha-globulin could be integrated here at their corresponding place in the ontology (see below). I recommend doing this. Furthermore, it might be sensible to add all the individual proteins that belong to one of the three different groups (alpha-, beta-, and gamma-globulins) as subclasses, e.g. HP:0032025 Reduced circulating alpha-1-antitrypsin concentration as subclass to the (new) class "Decreased circulating alpha-globulin concentration". The latter should be subclass to the (new) class "Abnormal circulating alpha-globulin concentration". I recommend doing this as well. (See also Appendix below)	
HP:0040228	Decreased level of plasminogen	Plasminogen belongs to the group of beta-globulins. Therefore HP:0040228 (new label: "Decreased circulating plasminogen concentration") should be subclass to the (new) class "Decreased circulating beta-globulin concentration" (→ subclass to HP:0025465 Abnormal circulating beta globulin level, see below).	reclassify
HP:0032312	Decreased circulating globulin level	I recommend eliminating this class for the sake of simplicity and clarity. But if this class is to be retained as subclass to HP:0010876 Abnormal circulating protein concentration, it should be sibling class to the three (new) classes "Abnormal circulating alpha-/beta-/gamma-globulin concentration". In this case the three (new) classes "Decreased circulating alpha-/beta-/gamma-globulin concentration" (with corresponding child classes) should be added as subclasses to HP:0032312 Decreased circulating globulin level.	eliminate
HP:0032311	Increased circulating globulin level	I recommend eliminating this class for the sake of simplicity and clarity.	eliminate

		But if this class is to be retained as subclass to HP:0010876 Abnormal circulating protein concentration, it should be sibling class to the three (new) classes “Abnormal circulating alpha-/beta-/gamma-globulin concentration”. In this case the three (new) classes “Elevated circulating alpha-/beta-/gamma-globulin concentration” (with corresponding child classes) should be added as subclasses to HP:0032311 Increased circulating globulin level.	
HP:0005413	Increased alpha-globulin	This class should be renamed (“Elevated circulating alpha-globulin concentration”) and classified as subclass to the (new) class “Abnormal circulating alpha-globulin concentration” (see above).	rename/ reclassify
HP:0025465	Abnormal circulating beta globulin level	This class should remain subclass to HP:0032179 Abnormal circulating globulin concentration. Analogously, two (new) classes “Abnormal circulating alpha-globulin concentration” and “Abnormal circulating gamma-globulin concentration” should be added as sibling classes. HP:0025465 should be renamed: “Abnormal circulating beta-globulin concentration” (please add hyphen).	add/ rename
HP:0025465	Abnormal circulating beta globulin level	The already existing subclasses of HP:0025465 Abnormal circulating beta globulin level (→ “Abnormal angiostatin/properdin/transferrin/sex-hormone binding globulin level”) should each be supplemented with the corresponding (new) subclasses “Elevated/Decreased circulating angiostatin/properdin/transferrin/sex-hormone binding globulin concentration”. The already existing class HP:0031419 Reduced sex -hormone binding protein level (new label: “Decreased sex hormone-binding globulin concentration”) should be integrated at the appropriate place.	add/ integrate
HP:0410176	Abnormal circulating glucose-6-phosphate dehydrogenase concentration	The subclasses HP:0410180 Abnormal glucose-6-phosphate dehydrogenase level in dried blood spot, HP:0410183 Abnormal glucose-6-phosphate dehydrogenase level in leukocytes, HP:0410184 Abnormal glucose-6-phosphate dehydrogenase level in red blood cells and HP:0410185 Abnormal glucose-6-phosphate dehydrogenase level in tissue should be eliminated (with all corresponding subclasses), because they do not refer to circulating glucose-6-phosphate dehydrogenase concentration **in blood**.	eliminate

HP:0031031	Abnormal retinol-binding protein level	A new subclass “Elevated circulating retinol-binding protein concentration” should be added.	add
HP:0005559	Abnormality of the kinin-kallikrein system	If this complex class is to be retained, (at least) the following five proteins – each in the corresponding three variants Abnormal/Elevated/Decreased circulating … concentration – should be added as subclasses: Kallikrein, Prekallikrein, Kininogen, Bradykinin und Kallidin. The already existing classes HP:0034371 Reduced circulating prekallikrein concentration and HP:0005527 Reduced kininogen activity should be integrated (see below).	add
HP:0005527	Reduced kininogen activity	HP:0005527 Reduced kininogen activity (new label: “Decreased circulating kininogen concentration”) should be classified as subclass to the (new) class “Abnormal circulating kininogen concentration” (see above).	add
HP:0025498	Aceruloplasminemia	HP:0025498 Aceruloplasminemia should be classified as subclass to HP:0033144 Abnormal circulating ceruloplasmin concentration.	reclassify

Weitere Klassen

- Alle weiteren Klassen ab **HP:6000189 Decreased circulating ficolin 3 concentration** müssen um die jeweils fehlende Parent Class und Sibling Class ergänzt werden (Ausnahmen siehe Tabelle unten).
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    - **Beispiel**: HP:6000189 Decreased circulating ficolin 3 concentration muss ergänzt werden um “Abnormal circulating ficolin 3 concentration” und “Elevated circulating ficolin 3 concentration”.

Diesbezüglich sind folgende Besonderheiten zu vermerken:

HPO-ID	**Label**	**Comment**	**to do**
HP:6000968	Diminished circulating cationic trypsinogen concentration	The new parent class “Abnormal circulating cationic trypsinogen concentration” should be classified as subclass to the (new) class “Abnormal circulating trypsinogen concentration”, parent class to HP:0032493 Increased circulating trypsinogen (see below).	add/ reclassify
HP:6000485	Elevated circulating beta chorionic gonadotropin concentration	Parent and sibling class should be added → “Abnormal/Decreased circulating beta chorionic gonadotropin concentration”. I wonder why beta chorionic gonadotropin (= beta-hCG) is the only gonadotropin listed here as subclass to HP:0010876 Abnormal circulating protein concentration. There are three classes of gonadotropins: follicle-stimulating hormone (FSH), luteinizing hormone (LH), and human chorionic gonadotropin (hCG). All three are hormones (and glycoproteins), but only FSH and LH are classified as subclasses to HP:0030338 Abnormal circulating gonadotropin concentration (→ subclass to HP:0003117 Abnormal circulating hormone concentration). This should be reconsidered. Beta-hCG (= human chorionic gonadotropin) is a hormone that is mainly produced by the placenta during pregnancy. Therefore, the correlation between HP:6000485 Elevated circulating beta chorionic gonadotropin concentration and HP:0011433 High maternal circulating chorionic gonadotropin concentration (and HP:0011434 Low maternal circulating chorionic gonadotropin concentration) should also be checked. Question: Should HP:0011433 be subclass to HP:6000485 Elevated circulating beta chorionic gonadotropin concentration resp. HP:0011434 be subclass to the (new) class “Decreased circulating beta chorionic gonadotropin concentration”?	add/check
HP:0032640	Elevated circulating CCL18 level	Parent and sibling class should be added → “Abnormal/Decreased circulating CCL18 concentration”.	add/check

		I wonder why HP:6000446 Abnormal circulating CC chemokine concentration (with the subclasses HP:6000687 Elevated circulating CCL3 concentration and HP:6000688 Elevated circulating CCL4 concentration) is not classified as parent class to HP:0032640 Elevated circulating CCL18 level as well. Furthermore, I wonder why HP:6000446 Abnormal circulating CC chemokine concentration is not subclass to HP:0010876 Abnormal circulating protein concentration.	
HP:0003238	Hyperpepsinogenemia I	Hyperpepsinogenemia (= "Elevated circulating pepsinogen concentration") is a medical term that can be retained. Parent Class ("Abnormal circulating pepsinogen concentration") and Sibling Class ("Decreased circulating pepsinogen concentration" or „Hypopepsinogemia") should be added.	add
HP:0002152	Hyperproteinemia	I recommend eliminating this class for the sake of simplicity and clarity. Should this class be retained, all classes following the scheme "Elevated circulating … concentration" should be subsumed under HP:0002152.	eliminate
HP:0003075	Hypoproteinemia	I recommend eliminating this class for the sake of simplicity and clarity. Should this class be retained, all classes following the scheme "Decreased circulating … concentration" should be subsumed under HP:0003075.	eliminate
HP:0032493	Increased circulating trypsinogen	The new parent class of HP:0032493 ("Abnormal circulating trypsinogen concentration") should also be parent to the new class "Abnormal circulating cationic trypsinogen concentration" (parent class to HP:6000968 Diminished circulating cationic trypsinogen concentration, see above). The sibling classes should be classified accordingly.	add/ reclassify
HP:0032025	Reduced circulating alpha-1-antitrypsin	HP:0032025 (new label: "Decreased circulating alpha-1-antitrypsin concentration") should be added as subclass to the (new) class "Decreased circulating alpha-globulin concentration". The corresponding parent and sibling classes "Abnormal/Elevated circulating alpha-1-antitrypsin concentration" should be classified accordingly as subclasses to "Abnormal/Elevated circulating alpha-globulin concentration" (see above).	rename/ reclassify

HP:0034371	Reduced circulating prekallikrein concentration	HP:0034371 should be subclass to the new class “Abnormal circulating prekallikrein concentration” (→ subclass to HP:0005559 Abnormality of the kinin-kallikrein system, see above).	reclassify
HP:0031419	Reduced sex -hormone binding protein level	HP:0031419 should be renamed (”Decreased circulating sex hormone-binding protein concentration”) and be integrated as subclass to its (already existing) parent class HP:0031438 Abnormal sex hormone-binding globulin level (new label: ”Abnormal circulating sex hormone-binding protein concentration”). Sibling class (”Elevated circulating sex hormone-binding protein concentration”) should be added.	rename/ reclassify/ add

Kommentare

- Die erforderliche Umbenennung der Label, z. B. Ersatz von „reduced“ durch „decreased“ oder „level“ durch „concentration“, habe ich bis auf ein paar Ausnahmen nicht angemerkt, da ich von einer automatischen Umsetzung im Vorfeld ausgehe.
- Wenn eine Klasse aus meiner Sicht nicht als Subklasse für HP:0010876 Abnormal circulating protein concentration geeignet ist, habe ich dies mit dem Vermerk „eliminate“ versehen. Das schließt nicht aus, dass die betroffene Klasse an einer Stelle der gesamten Ontologie richtig platziert werden kann (das habe ich nicht untersucht).
- Bei einigen Proteinen wird auch das völlige Fehlen der entsprechenden Substanz im Blut als eigene Klasse kodiert, z. B. HP:0025498 Aceruloplasminemia (= völliger Mangel an Ceruloplasmin). Hier sollte ggf. darüber nachgedacht werden, ob ein solcher Fall nicht auch bei anderen Substanzen auftreten kann und dann entsprechend auch kodiert werden sollte. Zudem stellt sich die Frage, ob das völlige Fehlen eines Proteins als Sonderfall einer reduzierten Konzentration angesehen werden sollte. In diesem Fall müsste eine solche Klasse entsprechend als Subklasse eingeordnet werden (z. B. HP:0025498 Aceruloplasminemia als Subklasse von „Decreased circulating ceruloplasmin concentration“).
- Grundsätzlich sollte aus meiner Sicht die Entscheidung getroffen werden, an welcher Stelle einzelne Proteine, die übergeordneten Gruppen angehören (z. B. Globuline), eingeordnet werden sollen (z. B. die neue Klasse „Abnormal circulating alpha-1-antitrypsin concentration“ als direkte Subklasse von HP:0010876 Abnormal circulating protein concentration und/oder als Subklasse der neuen Klasse „Abnormal circulating alpha-globulin concentration“, ihrerseits Subklasse zu HP:0032179 Abnormal circulating globulin concentration.

Letztlich müssten alle Subklassen von HP:0010876 Abnormal circulating protein concentration, die sich auf einzelne Proteine beziehen, daraufhin untersucht werden, ob es sich um Alpha-, Beta- oder Gammaglobuline handelt, und diese müssten dann an der entsprechenden Stelle in die Ontologie eingefügt werden. Gegebenenfalls könnte auch noch eine Unterteilung in Alpha-1- und Alpha-2-Globuline bzw. Beta-1- und Beta-2-Globuline getroffen werden.

Appendix: List of globulins (not complete)

Globulins	**Protein**	**ID**	**Label**
Alpha-globulins	Alpha-1-Antitrypsin	HP:0032025	Reduced circulating alpha-1-antitrypsin concentration
	Haptoglobin	HP:0020179	Abnormal circulating haptoglobin concentration
	Ceruloplasmin	HP:0033144	Abnormal circulating ceruloplasmin concentration
	Thyroxine-binding globulin	HP:0034590	Abnormal circulating thyroxine-binding globulin concentration
	Retinol-binding protein	HP:0031031	Abnormal retinol-binding protein level
Beta-globulins		HP:0025465	Abnormal circulating beta globulin level
	C-reactive protein (CRP)	HP:0032436	Abnormal circulating C-reactive protein concentration
Gamma-globulins	[gamma-globulins = immunoglobulins]	HP:0010701	Abnormal circulating immunoglobulin concentration